REVERSE

GRAY HAIR

Nutrient Complex

Single-bottle nutritional supplement / one proposed daily adult serving



Updated packaging concept: one bottle and carton, PEPTIDE SERIES branding and nutrient highlights. The 60-capsule pack is a design target; serving and fill specifications require manufacturer confirmation.

ONE BOTTLE / NINE-NUTRIENT FORMULA

Nutritional support proposal. Gray-hair efficacy remains unestablished.

Recommended nutrient formula

Code VTD-RGH-N01 / target one daily adult serving. Final capsule count and fill depend on the chosen grades.

Nutrient	Daily amount	Proposed source	Manufacturing basis
Copper	1 mg elemental	Copper gluconate	Assay the elemental copper
Zinc	5 mg elemental	Zinc citrate	Assay the elemental zinc
Vitamin B12	25 mcg	Cyanocobalamin	Use a validated premix
Folate	200 mcg folic acid	Folic acid	About 340 mcg DFE with food
Vitamin B6	1.4 mg	Pyridoxine hydrochloride	Amount as vitamin B6
Vitamin D3	15 mcg / 600 IU	Cholecalciferol	Confirm premix potency
Vitamin C	80 mg	Ascorbic acid	Amount as vitamin C
Pantothenic acid	5 mg	Calcium D-pantothenate	Amount as vitamin B5
Biotin	50 mcg	D-biotin	Use a validated premix

Serving and pack proposal

Aim for one HPMC capsule per daily serving if assay-corrected powder weights and fill density allow. A 30-serving or 60-serving single bottle can be selected after development. These are nutrient-delivery targets, not a complete raw-powder bill of materials.

Why this formulation: modest nutrient quantities avoid building the concept around high-dose supplementation. The amounts are proposed nutrition levels, not demonstrated gray-reversal doses. Adding B5, vitamin C and low-dose biotin broadens the nutritional profile; superior gray-hair results have not been established. [4] [5] [6] [7] [8]

Turn nutrient targets into a batch

Use actual supplier assays and legal ingredient forms. Do not weigh the delivered nutrient amounts as if they were salt or premix weights.

Calculate raw-material quantities

Raw-material mass = target nutrient mass divided by the assayed nutrient fraction. For copper and zinc, use elemental content. For calcium D-pantothenate, use vitamin B5 content. Include the carriers of all premixes in total fill mass; do not add a separate calcium target from the B5 salt.

Two examples - illustrative assays only

B12: a hypothetical 1.00% B12 premix requires 2.50 mg premix per serving to deliver 25 mcg B12.
D3: a hypothetical 100,000 IU/g premix requires 6.00 mg per serving to deliver 600 IU.
Use these only to understand the calculation. Actual purchase grades and assays must determine the batch sheet.

Establish blend uniformity and capsule fit

Use validated premixes and a dilution/blending strategy suitable for microgram nutrients. Choose excipients after bulk-density and flow assessment; verify blend homogeneity and capsule weight variation. Define processing humidity, moisture protection and an appropriate container-closure system from development data.

Confirm compatibility and shelf life

Screen the complete mineral/vitamin mixture rather than assuming it is stable. Consider protected or separately granulated grades if needed, especially when formulating minerals with vitamin C and B12. Set justified overages from stability data and applicable limits; avoid automatic high overages. Verify nutrient content at release and through the claimed shelf life.

What the supplement can support

This proposed capsule is nutritional support, not a proven treatment for established gray hair or a diagnosed deficiency.

Evidence boundary

The 2026 Arey study tested a different finished supplement and serum in four arms over 24 weeks. Its supplement included a proprietary Fo-Ti/PABA/sesame blend and other nutrients. Results cannot establish efficacy of this new VETUD capsule or identify which individual ingredient caused an effect. [1]

Why Fo-Ti and catalase are excluded

Polygonum multiflorum (Fo-Ti) has documented associations with clinically apparent liver injury. Catalase formulation research addresses gastrointestinal protection; it does not demonstrate human gray-hair repigmentation. Neither ingredient is included in this proposal. Iron is also absent because supplementation should depend on an identified need. [3] [2]

Adult suitability and review

Assess total nutrient intake from other supplements and diet. Establish appropriate label exclusions and professional-review advice, including for copper-metabolism disorders, relevant medicines, pregnancy or breastfeeding. This formula is not intended for children and should not replace clinical evaluation of sudden or unusually early graying. [4] [5] [6]

Release and claims work

Verify raw-material identity, nutrient assays, contaminants, microbial quality, capsule performance and container protection through a qualified supplement manufacturer. Confirm permitted forms, daily levels and market-specific labeling. PEPTIDE SERIES is a brand line; this nutrient formula contains no peptide actives. The working name "Reverse Gray Hair" is an efficacy claim requiring substantiation and review.

What the formulator should return

One finished formula for one bottle. This document does not propose a second enzyme tablet or a second daily supplement.

Requested deliverables

Return the assay-corrected ingredient weights per serving and batch, supplier-grade specifications, premix carriers, capsule size and count, excipient quantities, uniformity method, nutrient overage rationale, stability and testing plan, preliminary label values and a costed single-bottle packaging proposal. Identify unresolved safety, compatibility or efficacy questions.

[1] Bhardwaj, V., et al. (2026).
Randomized double-blind, placebo-controlled study of a supplement and serum on grey hair.
Journal paper / DOI

[2] Alothman, M., et al. (2021).
Design of catalase monolithic tablets for intestinal targeted delivery. Pharmaceutics, 13, 69.
Journal paper / DOI

[3] NIDDK / LiverTox.
Polygonum multiflorum: documented liver-injury concerns.
NIH safety reference

[4] NIH Office of Dietary Supplements.
Copper: nutritional needs, excess intake and special considerations.
Copper fact sheet

[5] NIH Office of Dietary Supplements.
Zinc: nutrient intake, safety and interactions.
Zinc fact sheet

[6] NIH Office of Dietary Supplements.
Vitamin B12: nutrient status and supplemental forms.
Vitamin B12 fact sheet

[7] NIH Office of Dietary Supplements.
Folate: folic acid, dietary folate equivalents and intake references.
Folate fact sheet

[8] NIH Office of Dietary Supplements.
Vitamin D: intake and safety references.
Vitamin D fact sheet

Document status: development proposal prepared for VETUD, 09 November 2026. Proposed nutrient amounts require qualified formulation and suitability review. Clinical gray-hair benefit of this capsule and its combination with the serum has not been established.